



April 10, 2022

Dr. Stephanie Johnson
U.S. Department of Energy
Office of Energy Efficiency and Renewable Energy
Building Technology Office, EE-5B
1000 Independence Avenue, SW
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Submitted via www.regulations.gov

Re: Blueair IAQ Comments on DOE's Request for Information for the Test Procedure and Energy Conservation Standard for Consumer Air Cleaners [Docket ID EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036]

Dear Dr. Johnson:

Blueair respectfully submits the following comments to the Department of Energy (DOE or Department) on its Request for Information on the Test Procedure and Energy Conservation Standards for Consumer Air Cleaners (Air Cleaner RFI); Docket Nos. EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036; Fed. Reg. 3702 (Jan. 25, 2022)

Blueair began with a commitment to provide products that would enhance respiratory health and relieve the discomfort of allergy and asthma symptoms. We actively research new clean air solutions for homes, offices, and commercial locations. Customers choose our products because of their reputation for being cost-effective, reliable, and long-lasting. In addition, Blueair's products include solutions for in-home applications to professional settings.

Blueair would like to note our appreciation for the open process with which the Department seeks input from stakeholders and industry.

Blueair supports DOE's efforts to save energy and ensure a national marketplace through the Appliance Standards Program. We continue to support DOE's proposal to include consumer air cleaners as covered products. The implementation of a Federal standard is a welcomed development in the marketplace. As a manufacturer, we focus on putting high-quality, effective products in our consumers' hands that they can be confident will achieve the outcomes we advertise. We believe that educated consumers make better decisions and that adopting a Federal testing standard will increase transparency and ensure access to high-quality energy-efficient products.



I. Scope and Definition

Blueair agrees with DOE's proposal to include consumer air cleaners in its scope of coverage and to consider possible test procedures and energy conservation standards for these products. Blueair has no recommended changes to the rule as proposed below:

DOE proposed in the September 2021 NOPD to define a consumer air cleaner as a consumer product that:

- (1) Is a self-contained, mechanically encased assembly;
- (2) Is powered by single-phase electric current;
- (3) Removes, destroys, or deactivates particulates and microorganisms from the air; and
- (4) Excludes products that destroy or deactivate particulates and microorganisms solely by means of ultraviolet ("UV") light without a fan for air circulation; and
- (5) Excludes central air conditioners, room air conditioners, portable air conditioners, dehumidifiers, and furnaces as defined in 10 CFR 430.2.. 86 FR 51629, 51632.

As discussed in the September 2021 NOPD, DOE proposed to exclude from coverage those consumer products that purify air solely by means of UV light without circulating air through the product using a fan because the energy-consuming component of such products would be a fluorescent lamp or light-emitting diode designed to emit light in the UV portion of the electromagnetic spectrum. 86 FR 51629, 51632. Accordingly, DOE would classify these products under EPCA as a type of lamp (see the definition of "lamps primarily designed to produce radiation in the ultraviolet region of the spectrum" and "light-emitting diode or LED" in 10 CFR 430.2), and therefore, did not consider applying any future consumer air cleaner requirements to these products. *Id.*

DOE continues to evaluate comments received from interested parties in response to the proposed definition for consumer air cleaners in the September 2021 NOPD.

II. Test Procedure For Consumer Air Cleaners

DOE requested comments on the appropriate test method for measuring room air cleaner efficiency. Blueair believes that the ANSI/AHAM AC-1-2020 testing and calculation metric of smoke clean air delivery rate per watt is the most robust way to determine air purifier energy efficiency. ANSI/AHAM AC-1-2020 is a well-established testing method in the United States. In addition, there is a steady supply of laboratories across the country that can readily run tests for manufacturers and third parties at reasonable costs and turnaround times.

Blueair believes that the ANSI/AHAM AC-1-2020 testing is appropriate for defining energy conservation standards for consumer air cleaners and should serve as the Federal test. The output of product



performance (clean air delivery rate) per watt is similar to how the lighting industry energy efficiency uses lumens per watt as the metric for energy efficiency in their products.

Blueair maintains that the clean air deliver rate (CADR) is an essential and useful tool for comparing the overall performance of different models, sizes, and shapes of consumer air cleaners. The Environmental Protection Agency highlights, “it is crucial to understand the difference between two parameters that influence the performance of air-cleaning devices: efficiency and effectiveness. The efficiency of an air-cleaning device is a fractional measure of its ability to reduce the concentration of pollutants in the air that passes once through the device. The fractional efficiency is measured in a laboratory, where all relevant variables are controlled. The effectiveness of an air-cleaning device or system is a measure of its ability to remove pollutants from the spaces it serves in real-world situations. Therefore, the most helpful parameter for understanding the effectiveness of portable air cleaners is the CADR, which is a measure of a portable air cleaner’s delivery of relatively clean air, expressed in cubic feet per minute (cfm). The CADR is a product of the fractional removal efficiency for a particular pollutant and the airflow rate through the air cleaner”.

Blueair supports the adoption of a metric based on a PM 2.5 Clean Air Deliver Rate (CADR). PM 2.5, represented by a combination of measured smoke and dust particle sizes, measures the rate at which a room air cleaner removes fine particulate matter from the air. Fine particulate matter has been shown to cause serious health problems and can get into the lungs and bloodstream. The Environmental Protection Agency (EPA), among other experts, has indicated that PM 2.5 is the primary concern from a health standpoint and is often found indoors. Accordingly, this particulate is likely to be of greatest concern to consumers. It is also a very fine particulate, and, thus, it can adequately represent a unit’s performance for other particles as well. AHAM AC-7, in its current draft form, measures efficiency based on PM 2.5 as the numerator.

III. Metric For Consumer Air Cleaners

DOE requested comment on consumer usage of consumer air cleaners. Blueair advocates for the continued use of the ENERGY STAR program, which assumes 16 active mode hours per day and 8 inactive mode hours per day to calculate the annual energy consumption of qualifying consumer air cleaners.

DOE requested comment on whether cigarette smoke would be the appropriate particulate for determining the CADR rating of air cleaners under a DOE test procedure. Blueair is supportive of the continued use of cigarette smoke as the appropriate particulate for CADR ratings as it can be a surrogate for much smaller particles that can be found in the home. However, Blueair maintains that any pollutants specified in ANSI/AHAM AC-1-2020 would be suitable.

DOE requested comment on the technical feasibility of integrating measures related to overall energy consumption and energy efficiency. For example, Blueair’s trademarked HEPASilent technology uses a combination of electrostatic and mechanical filtration to remove 99.97% of airborne particles down to 0.1 microns. Blueair’s HEPASilent technology is a significant innovation that proves the consumer air



cleaner space can provide a higher particle CADR using less energy. Yet the widespread marketing of HEPA, referring only to a filter's filtration efficiency without consideration of the air purifier's/system effectiveness of airflow rate Blueair deems as an insufficient metric for evaluating the full performance of the air purifying device (per EPA's highlight). Hence, Blueair continues to support the EPA Energy Star rating for an air purifier that considers CADR per watt as the metric that measures energy efficiency for performance output. Blueair's position is that technology will continue developing and demonstrating additional energy conservation capabilities through proven and aligned testing, none of which are cost-prohibitive or scarce.

IV. Potential Energy Conservation Standards

DOE requested comment on whether energy conservation standards for consumer air cleaners would be economically justified, technologically feasible, and would result in a significant savings of energy. Blueair supports the implementation of a conservation standard. The pandemic and natural disasters such as the recurring wildfires in regions across the country have highlighted the need for consumer air cleaners in many private and public indoor spaces to manage air quality and support occupant well-being. Blueair's HEPASilent technology has reduced energy consumption for maximum clean air delivery on a mass scale. In addition, it has demonstrated that it is technologically possible to design and manufacture air purifiers with reduced energy usage without loss of air cleaning performance.

Thank you for considering these comments.

Sincerely,

A handwritten signature in black ink, appearing to read "Daniel Espinosa".

Daniel Espinosa
Head of Strategy, IT, and Projects